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# A Transparency Statement Improves Community-Police Interactions

**Kyle Dobson**

kyle.dobson@utexas.edu

University of Texas at Austin

**Andrea Dittmann**

Emory University https://orcid.org/0000-0001-8250-0276

**David Yeager**

University of Texas at Austin https://orcid.org/0000-0002-8522-9503

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# Abstract

Over the last three decades, billions of dollars have been invested in community-oriented policing approaches that are meant to promote positive interactions between officers and civilians. However, trust in law enforcement remains low. Our qualitative analysis of over 500 hours of naturalistic observations suggested this is in part because police officers' manner of questioning can make civilians feel threatened, even when officers are not actively investigating a crime. Here we use a pre-registered field experiment ($N$ = 232 community members interacting with police officers) to show that civilians were less threatened and report greater trust when officers added a short "transparency statement" communicating their benevolent intent (e.g., "*I'm walking around trying to get to know the community*") to the beginning of interactions. Corroboration of these conclusions came from natural language processing analyses of the interaction transcripts and from ambulatory sensors that monitored community members' sympathetic nervous system arousal. Further, six online experiments showed that transparency statements were equally effective for members of historically-overpoliced groups (e.g., Black and Latinx civilians) and isolated the theoretical conditions under which transparency statements were most impactful (total $N$ = 3,180). This multi-method investigation highlights an under-appreciated reason why authority figures such as police so often fail to promote positive relationships with the community: a lack of transparency about the reasons for their behavior.


# Main Text

Since 1994, the U.S. federal government has invested more than $14 billion in more than 13,000 law enforcement agencies to support *community policing* efforts [1] . In community policing, officers are asked to prevent crimes by centering decision-making within the community. Officers question community members, solve community problems, and cultivate cooperative relationships, even when they are not actively interrogating a suspect in a potential crime [2,3] . By 2013, 90% of medium-to-large police departments in the U.S. claimed to be using community policing to at least some extent, but these approaches remain unstandardized [4] .

Community policing, on its face, appears to be a more respectful approach than the *proactive policing* model. In contrast to community policing, proactive policing typically involves an "investigatory stop," in which police intrusively question people (e.g., "What are you doing here?"), pat them down, search them, or handcuff them when suspected of illicit activity[5]. One well-known manifestation of this is the "stop and frisk" policy adopted in New York City in the 1990s[6]. Proactive policing is a dominant approach to reducing violent crime, but it has a high cost. It is a major contributor to the escalation of interactions that end in accusations of wrongful arrest or police brutality—particularly for historically over-policed communities of color and those from lower-social-class backgrounds or neighborhoods[7].

Although community policing has the potential to repair trust between communities and law enforcement, the approach has so far been largely ineffective. Empirical evaluations show no average reduction in crime, and no increase in citizens' trust in the benevolence of police[8]. This is especially true

in historically over-policed communities, where trust in law enforcement is much lower on average[9]. Indeed, despite the prevalence of community policing initiatives, the public's trust as a whole in the legitimacy of law enforcement has still experienced a precipitous decline, and is now at an all-time low[10,11]. This loss of trust and legitimacy, in turn, makes citizens more hostile to police and less likely to comply with officers' requests for the information they need to solve a case—all of which make society less safe from harm[12].

Interestingly, the effects of strategies identified as falling under the umbrella of "community policing"—though not positive on average—are highly heterogeneous. A systematic review and meta-analysis[8] found that several studies showed significant benefits in line with policymakers' expectations[13]. However, several studies showed significant increases in crime and reductions in trust, contrary to the aims of the approach[14]. Furthermore, scholars have noted that law enforcement agencies claiming to implement community policing are often referring to very different practices [15]. This raises the possibility that it is not community policing itself, but rather something about how it is implemented, that accounts for the inconsistent results in prior evaluation studies.

Theorists of policing have suggested adopting *procedural justice* as a theoretical framework for improving community policing, but procedural justice has not yet made community policing more reliably effective[16–18]. Procedural justice theory emphasizes that community members' trust of police officers comes from their assessment of whether an authority figure is legitimate, and these assessments of legitimacy come from being treated with dignity, respect, and politeness[19,20]. Although these claims are well-supported in basic research, when procedural justice treatments have been implemented in real-world settings it has led to mixed results [17,18,21] and a high degree of unexplained heterogeneity (Average $I^2$ = 95%)[16]. Systematic reviews have suggested that this heterogeneity may be the result of procedural justice trainings allowing a great deal of variation in individual officers' choices of how to implement the theory's recommendation.

Here we present a method for improving the reliability of trust-building interventions between police and communities that homes in on the specific behaviors (e.g., language) of officers that are consistent with the more abstract recommendations of procedural justice theory (e.g., being respectful and polite). We show that our proposed method is easy for police officers to implement with little training, and that it reliably decreases community members' sense of threat and increases their trust in live interactions. Moreover, our results suggest that our proposed method has benefits even for those members of groups that have been historically over-policed (i.e., people of color and those from lower-social-class contexts).

Our research begins with the observation that officers have discretion over their conversational dynamics, but they often fail to leverage that discretion in a way that satisfies their goals of building trust. A qualitative study of police officers during ride-alongs (i.e., routine officer patrols attended by researchers[22]) found that civilians with whom officers were interacting worried that they were being approached for enforcement reasons. In part, this was because patrol officers continued to use

aggressive questioning methods that were tailored for suspects (e.g., “What are you doing here? How long have you been here? Where are you going next?”)—even when officers were attempting to cultivate trusting relationships with community members. That study[22] furthermore found that community members perceived officers’ interest in them as harassment. As a result, community members anxiously tried to disengage from the conversation from the start, only to feel increasingly threatened as the interaction proceeded. At a more general level, community members have learned to fear that officers are accusing them of a crime and are likely to search or handcuff them during an interaction[22]. Officers in turn perceive attempts at disengagement or evasion as a suspicious sign that the community member has something criminal to hide. This leads officers to escalate their tactics to interrogate or control community members, resulting in unnecessary arrests or physical harm. The result is a negative cycle of threat that undermines the goals of building trust, even when officers’ initial intent might have been positive.

Building on this prior work, we conducted ethnographic research to generate hypotheses about how to disrupt or even reverse this negative cycle. We conducted over 500 hours of field observations, interviews, and ride-alongs with police officers in law enforcement agencies in a diverse metropolitan area in the U.S. (see Box 1). We observed two units of police officers that reflected the heterogeneity in the effectiveness of community policing tactics[8]—one traditional patrol unit and another unit from a pilot program expressly developed to build relationships with the community. In both units, we observed interactions that were initiated with benevolent intent, in which officers were genuinely attempting to build relationships with their communities. However, the two units differed in the tactics they employed to achieve this goal. Some officers peppered community members with questions (e.g., “What are you guys doing here?”, “What are you up to?”), largely consistent with proactive policing tactics[5]. When we interviewed community members after those interactions, citizens described feeling threatened, uncertain if they were going to be accused of a crime, and confused as to why the officers chose to talk with them (see left panel in Box 1). Other officers, by contrast, were transparent. From the outset, they explained that their department had a new community policing initiative, and that their job was to get to know the community. Interactions between these officers and community members proceeded very differently. Community members appeared at ease and willing to talk with these officers (see right panel in Box 1; see SI for more details).

Our preliminary qualitative observations led us to generate a quantitative hypothesis about how to intervene to improve police-community interactions (see Fig. 1A). We theorized that a few words of clarification, in which the officer made was transparent that their questioning was not coming from a proactive policing model, but rather from a community policing model, might put the community member at ease and evoke positive engagement at the outset of the interaction. Importantly, because the first words in a conversation set the stage for everything that follows, if officers briefly clarified from the start what their positive goal is, it could prevent an escalating threat that has an outsized effect on the ultimate outcomes of the interaction.

We call this a *transparency statement*. It involves a clear and truthful statement of the officer's goals at the start of the interaction. Concretely, for example, it is the difference between an officer stating "Hi, I'm Officer [Name], how's it going? Can I talk to you for a minute?" which is not transparent but is the default for many officers, and an officer adding the ten-word statement "I'm walking around trying to get to know the community" before asking to talk with the person, which is transparent. See Fig. 1B. Note that without the transparency statement, the community member is free to assume that the officer's intentions are to repeat the past: aggressively interrogate a suspected criminal[23,24]. But with the transparency statement, the officer has presumably not left their intentions open to a negative alternative interpretation that could erode trust. To directly test this hypothesis, we conducted the present experiments.

**Box 1. Themes from Phase 1 that Yielded the Transparency Statement Hypothesis**

| Lack of Transparency Was Associated with<br><br>Threat and Fear of Being Accused of a Crime | Transparency Was Associated with<br><br>Positive Emotion and the Development of Rapport |
|---|---|
| Over a period of four weeks, we conducted four-hour ride-along observations with patrol officers in a metropolitan area in the Midwest three-to-four times per week. This was an area with high-profile accusations of police misconduct. We observed 18 total patrol officers interact with ~ 150 different civilians. Two key themes emerged:<br><br>Theme 1: Officers did not explain their intentions at the outset of the interaction. Officers frequently did not explain the reason for their interactions at the start of the conversation. Instead, intentions were only clarified toward the end, if at all. This was too late because, by this point, the civilians already appeared anxious and avoidant.<br><br>Ex 1: A White female officer drove through a city park, rolled down the window, and yelled at a civilian sitting on a bench:<br><br>Officer: "You okay?"<br><br>Civilian: "Yeah… why?" *Civilian stays seated, is standoffish, seems confused.*<br><br>Officer: "Just wanted to check—it's cold!" *Officer drives away with no further conversation*.<br><br>Theme 2: Officers' lack of transparency led civilians to wonder whether they were being accused of a crime. Although officers said that their interactions were received positively by the community, when we interviewed civilians, they felt confused by the conversations and were worried they would be accused of crimes.<br><br>Ex 2: Two White, male officers drove into a local park and approached four Black residents having a picnic. They began questioning the residents, saying "How are you?" and "What are you up to in the park today?" This line of questioning was met with terse answers, and did not lead to a back-and-forth rapport. After the officers left, the research team talked with the civilians. They described feeling anxious | Over a period of three years, we observed a new community policing unit within a large city in the Midwest which was known for accusations of police violence against community members. Within this pilot program, we observed 17 officers, four sergeants, one lieutenant, and one commander, for more than 500 hours. Two key themes emerged:<br><br>Theme 1. Officers proactively made their intentions transparent immediately, at the outset of the interaction. The officers in this community policing unit were aware that their very presence could make civilians anxious and put them on edge. Therefore, they proactively explained why they were approaching them, before they said anything else.<br><br>Ex 1: A Latino male officer was approaching a civilian who was involved in prostitution. This is an arrestable offense but the officer was not, at that time, planning to arrest the civilian. Instead the officer's goal was to clean up a certain corner in the neighborhood. The officer said: "I told them I wasn't arresting them, but 'I know what you're doing and you know what you're doing, and you shouldn't be doing that here.' After I got real with them, the person dropped their guard, calmed down, and got out of the area."<br><br>Theme 2. When officers began with a transparency statement, civilians appeared to feel positively about the police officers' actions, and a rapport was formed.<br><br>Often, the officers were attempting to do positive things for the community, such as look after a wayward young person. The officers were attuned to the fact that their interest in helping could be construed by the community as an attempt to criminalize or harass the young person. Therefore, they explained their intentions clearly. Then, when they offered help, it was often met with positive emotion and the development of rapport with the community members.<br><br>Ex 2: A Latina female officer and her partner had been talking with a local youth who needed help a few days previously. The officer described one day when they were on patrol and "we saw his aunts and uncles so we get down [from the car] and they're like '*Oh [expletive]!'* Immediately, right? We get the perception of like, 'The police are here. What happened?' So, we tell them, 'Hey, we're actually here to see [youth's name]. We met him a few days back and we wanted to bring him a little something. We promised we were |

**Box 1. Themes from Phase 1 that Yielded the Transparency Statement Hypothesis**

and unsure why the interaction was happening, and they said they felt threatened.

Ex 3: A Black male and an Asian-American male officer responded to a 311 call to investigate graffiti outside a building where there is a group therapy meeting going on. The officers walked around but did not explain their presence to anyone. The group therapy meeting attendees were distracted and staring at the uniformed officers. The meeting leader eventually approached the officers and said, "Yeah, we're having [a group therapy] meeting right now, so there are a bunch of [people] in there who are afraid you are coming for them."

going to come back and we just haven't been able to.' So [the aunts and uncles] are like, 'Really?' And it was really cool because one of the ladies, she was like, 'This is what [city] police should be about.'"

# Experiment With Live Interactions

Here we report a pre-registered, blinded, controlled field experiment that directly tested the causal effect of a transparency statement in live interactions between a community member and a uniformed officer. Note that this foundational experiment was critical for establishing the basic effects of transparency statements using precise experimental methods. As such, the experiment presented here were a precursor to a future program of research (not reported here) on the longer-term effects of transparency on public safety (which would afford less experimental control).

In Experiment 1, $N = 232$ adults who lived in a community known for recent accusations of officers' mistreatment and use of excessive force were told that we were studying natural interpersonal interactions and outfitted them with ambulatory physiological sensors. They were not told who they would interact with, or when. A few minutes later, participants were approached by a fully-uniformed officer carrying a weapon and asked questions for 1–2 minutes ($M_{age} = 22.07$, $SD_{age} = 5.90$; 54% female, 4% non-binary/other; 30% held a four-year college degree; 34% White/European American; 32% Asian/Asian American; 17% Latino/Hispanic American; 9% Black/African American; 8% Other). The study design is depicted in Fig. 1B. Officers were asked to have a conversation with a community member like they typically would when engaging in community policing activities. Half of the time the officer added a transparency statement (described above; see Box 2 for examples) when they introduced themselves, and half of the time they did not. The instructions given to officers did not differ in any other way and officers were not told the study hypotheses or outcomes.

**Overview of analyses.** We report raw means and standard deviations. Average treatment effects and posterior probabilities are calculated from Bayesian machine-learning regression models[25–27] that included covariates measured prior to assignment to condition. Inferences focused on estimation and

uncertainty, following best-practice recommendations[28]. All conventional null hypothesis tests for measures reported in the main text were $P < .05$ (see the SI).

**Manipulation check.** To measure transparency, community member participants answered our survey questions about the interaction immediately after it ended (See Fig. 1B). They rated the officer's transparency (three items, e.g., "The officer I talked to stated their intentions for talking to me clearly"). As expected, this analysis showed that community members rated the officer as more transparent in the transparency condition ($M = 5.64$, $SD = 1.21$) relative to the control condition ($M = 4.40$, $SD = 1.61$), average treatment effect (ATE) = .77 *SD* [posterior distribution 10th %ile = 0.50, 90th %ile = 1.01], posterior probability that the (ATE > 0) > .99. See Fig. 1C. Note that officers chose to implement this in idiosyncratic ways due to the same discretion they would have outside of the field experiment (see Box 2). Nevertheless, officers' introductory statements in the treatment condition were more transparent, compared to the control condition.

**Box 2. Examples of Transparency Statements in Experiment 1**

In Experiment 1, officers were given discretion in delivering transparency statements, similar to how they would outside the experimental context. The similarities in their statements is reflected in the fact that they start by indicating their benevolent intent for initiating the interaction, and in a meaningful manipulation check effect (see the main text).

• "Hello, I'm just out and about walking around, talking to people in the community. Is it okay if I talk to you for a minute?"

• "Hey, my name is [Name], and I'm just taking a walk, trying to get to know my community better. You mind if I sit and talk to you for a second?"

• "I'm [Name] with [Department] and I'm just walking around getting to know like everybody that's hanging out in the area. [To] just introduce myself."

**Pre-registered outcomes.** The primary pre-registered outcome was community members' perceived *threat* from law enforcement (3 items, e.g. "*The police officer did not trust you*," "*The officer just wanted to make friendly conversation*" (reverse-scored)). Community members reported feeling less threatened in the transparency condition ($M = 2.92$, $SD = 1.45$) relative to the control condition ($M = 3.40$, $SD = 1.41$), ATE = − .20 *SD* [-.43, − .01], pr(ATE < 0) = .92.

The secondary pre-registered outcome was *trust* in the officer's benevolent intentions (2 items,

"*In the future, how likely is it that this police officer would go out of their way to help you*," "*In the future, how likely is it that this police officer would care about you and your welfare*," 1 = *Very unlikely*, 7 = *Very likely*). As predicted, community members reported greater trust in the transparency condition ($M = 5.66$, $SD = 1.14$) relative to the control condition ($M = 5.31$, $SD = 1.28$), ATE = 0.17 [0.001, 0.38], pr(ATE > 0) = .90. See Fig. 1B.

**Language and emotion during the interaction.** How did transparency affect the unfolding of the interactions over time? To answer this, a principled exploratory analysis was conducted. This involved a)

statistical and idiographic analysis of the language in the transcripts, and b) use of conservative, Bayesian models that reduce false-positive results.

First, natural language processing algorithms[29] were applied to the community members' conversation transcripts. A statistical analysis of all of the conversation transcripts and reflections was conducted, utilizing the popular and well-validated Linguistic Inquiry & Word Count (LIWC-22) software[29], which identifies linguistic features of transcripts of which most people are not conscious. Transcripts from the transparency condition showed that the community members used more *authentic* language by the end of the interaction (i.e. the final third, in terms of each transcript's word count), ATE = 46 *SD* [.23, .66], pr(ATE > 0) = .99. Authentic language refers to language in which people reveal themselves in a genuine or honest way, not censoring or filtering. This difference across conditions is meaningful because it reveals that participants were using the type of language that forms the foundation of positive rapport that makes people feel unthreatened and open to trust. An analysis of word count also found that community members talked more with officers in the transparency condition, ATE = 175 words [129, 232], pr(ATE > 0) = .99, consistent with the notion that transparency helped officers establish rapport.

On the post-experimental survey, participants reflected on thoughts and feelings from the end of the interaction. In the transparency condition civilians' text included less *causal language*, ATE = − .27 *SD* [-.48, − .08], pr(ATE < 0) = .97, suggesting transparency reduced participants' puzzlement. The natural language analysis also found somewhat more positive emotion words in community members' written reflections, ATE = .14 *SD* [-.008, .32], pr(ATE > 0) = .88. Finally, participants directly reported on the survey which emotions they felt at the end of the interaction with the officer. Participants in the transparency condition were less than half as likely to report threatened emotions (afraid, nervous, upset, hostile, ashamed) relative to the control condition (Raw data Control = 38%; Transparency = 19%), ATE = − .15pp [-.25, − .05], pr(ATE < 0) = .975.

These statistical patterns were supported by an idiographic analysis of participants who were identified as statistical outliers with respect to the two pre-registered outcomes of threat and trust[30]. Idiographic analyses in Fig. 2, Panels A-B, show how different the conversations were in the absence versus presence of a transparency statement. On the top (Panel A), the civilian and officer never established a rapport. Instead, we see terse answers. The civilian is guarded and self-conscious. This interpretation is confirmed when we read their reflections after the interaction. The participant reported being puzzled about why the officer was approaching them and concerned by the threatening prospect that they were being investigated for a crime.

At the bottom of Fig. 2 (Panel B), the interaction in the transparency condition is quite different. When the officer begins the conversation with a transparency statement, it ends with the spontaneous back-and-forth, almost resembling new friends who are at ease with one another. In fact, the civilian invites the officer to have future interactions with them at their place of work. When the civilian reflected on this interaction, they admitted that although they were apprehensive at the very start, by the end they felt

“comfortable and like we were friends.” Overall, these transcripts matched the trends we saw in our qualitative research (Box 1).

**Stress physiology.** We next explored whether the colder and more avoidant response from civilians in the control condition and the more active, warm, engaged response of the transparency condition could be detected through an analysis of sympathetic nervous system (SNS) responses. We conducted an exploratory analysis of skin conductance levels (SCL), obtained from electrodermal activity (EDA) sensors embedded in a wrist-worn ambulatory device. These sensors captured *in vivo*, temporal dynamics of the stress responses during community members’ interactions with police officers. Higher SCLs have sometimes been viewed as evidence of negative stress, but in the present case they were likely to be positive signs of arousal and active engagement in the conversation. This expectation was derived from the Biopsychosocial (BPS) model of challenge and threat[31–33]. In addition, to validate the ambulatory EDA data processing algorithms and interpretations, we conducted a pilot laboratory study ($N$ = 100) that administered a controlled stressor (the Trier Social Stress Test) and collected data from the same wearable EDA sensors as in the field study, along with gold-standard psychophysiological measures of challenge and threat (e.g. Total Peripheral Resistance; details reported in SI Fig. S2A-B). Based on the results of our prior validation study, we expected the following trends to emerge when participants received the transparency statement: (a) initial effects on higher SCL at the outset of the interaction relative to controls, and (b) quicker recovery to homeostasis over time, resulting in negative/reversed treatment effects relative to controls during the recovery epoch.

Indeed, we found a positive effect of transparency statements on SCL during the initial ten seconds of the interaction, CATE = 0.13 *SD* [0.04, 0.25], pr(ATE > 0) = .97. See Fig. 3A and B. This shows greater arousal, and it is consistent with openness to a new, positive relationship with the police officer. Next, we found a stronger decline over time in SCL in the transparency condition, consistent with less arousal. By the end of the interaction, the treatment effect had reversed, yielding a difference in CATEs for the first 10 seconds vs. the recovery period of .21 *SD* [.08, .36], pr(CATE for first 10 seconds > CATE for recovery) = .99. In summary, the EDA data in Fig. 3A and B were consistent with greater initial arousal, and less arousal during recovery, in the transparency condition compared to the control condition.

## Replication And Moderation By Racial/ethnic Demographics

Experiment 2 ($N$ = 969) was a pre-registered online study that replicated Experiment 1’s findings using a brief video-recorded introduction from a uniformed police officer stopping to talk with a community member in a park. The sample was constructed to resemble the U.S. population’s marginal distributions of age, gender, and race/ethnicity. Because a key goal of community (vs. proactive) policing is to reduce inequality in trust in police, this study tested whether group memberships moderated the effects of transparency. On the one hand, people of color and those from lower-SES contexts might have more negative attitudes towards law enforcement at baseline, potentially reducing the efficacy of a simple transparency statement[9]. On the other hand, over-policed racial, ethnic, and socioeconomic groups

might benefit the *most* from transparency because it addresses the attributional ambiguity that comes from being targeted by stereotypes[23,24]. The present study had the opportunity to assess these two competing hypotheses using a sample with greater statistical power than Experiment 1. A secondary goal of Experiment 2 was to replicate our primary results using a more comprehensive, 10-item measure of threat.

We collaborated with a White, male police officer to co-create an authentic implementation of the transparency statement in real-world settings for this video manipulation. In both conditions, we used the same White, male focal officer for two reasons. First, a White, male officer is the prototypical police officer in the United States. From the most recent statistics in 2020, the population of police officers in the US comprised 86% men and 66% non-Hispanic White officers.[36] Second, White, male police officers are likely perceived as most threatening compared to all other racial and gender groups because they are the most likely to use force against community members.[37] In the control video, the police officer said "Hey there, I'm [Officer Name], I'm just out here saying hi and seeing how things are going" (cf. Box 2). In the transparency condition, the officer said, "I'm out here on foot patrol, just trying to get to know my beat and my community." The specific wording of this transparency statement was developed to feel authentic to the officer, resulting in greater experimental realism[38], but it nevertheless kept to the core of a transparency statement, which was to explain the reasons for their behaviors.

Replicating Experiment 1, the transparency statement reduced threat on the full 10-item measure: ATE = − .28 [-.48, − .11], pr(ATE < 0) = .988. Results were the same for the 3-item subset (matching Experiment 1): ATE = − .42 [-.65, − .22], pr(ATE < 0) = .996. The transparency statement also increased trust, ATE = .13 [.005, .28], pr(ATE > 0) = .912, and behavioral intentions to initiate rapport with officers in the future (focal officer: ATE = .47 [.21, .75], pr(ATE > 0) = .995; other officers from the same department: ATE = .16 [.02, .32], pr(ATE > 0) = .936). Thus the transparency effects replicated in this national, diverse sample.

Interestingly, we found no consistent moderation by demographic group (i.e., age, gender, race/ethnicity, native U.S. status, education, income, conservatism, baseline attitudes towards police, and history of prior police contact; see Table S7 in the supplemental). In the few cases where there was moderation, the transparency statement was *more* effective for those who would be expected to have more negative attitudes towards the police initially (e.g., members of racial minority groups). This provides initial evidence that transparency statements reflect a scalable intervention with the potential to benefit individuals from historically over-policed communities.

## Controlled Online Experiments

What are the elements of a strong transparency statement, when do they work, and when do they not work? To answer these questions, we conducted five pre-registered, randomized, controlled experiments using online samples of adult volunteers who anticipated an interaction with an authority figure. In these experiments, we manipulated aspects of the introductory statements of the authority figures and

examined effects on the focal, pre-registered threat measure from Experiments 1 and 2. Extended details on the samples and study procedures are presented in the SI. An internal meta-analysis of all studies' results for the replication cells appears in Fig. 4.

In these five experiments, we moved to an online vignette-based design. We did so because our primary aim was to turn "on" and "off" certain elements of transparency statement using controlled experiments. After ensuring that the main effect of transparency statements translated to a vignette-based design, we varied aspects of the transparency situation to advance an understanding of mechanism.

An initial vignette study (Experiment 3, $N = 609$) replicated Experiment 1–2's findings using an anticipated interaction with a police officer stopping to talk with a community member outside of a grocery store. Participants in the control condition imagined that the police officer started a conversation by saying "Hi, how are you doing?" In the transparency condition, they anticipated interacting with an officer who explained that they were "just trying to get to know the community better." As with the live interaction and video study, the transparency statement reduced threat in this anticipated situation, ATE = -0.44 *SD* [-0.67, -0.22], pr(ATE < 0) = .99. In the four experiments that follow, the same effect was replicated each time (see the SI and Fig. 4).

We next asked: What is it about the statement of the officer's intentions that had a beneficial effect? In Experiment 4 ($N = 382$), we used a three-condition design to compare the transparency statement and the control condition to a new introduction that was more closely aligned with the aims of the aggressive policing approach (i.e. the officer is "trying to find a suspect in the area"). We suspected that the latter introductory statement would not be as effective because it does not disabuse the participant of the significant concern that they will be accused of a crime. We found that the aggressive policing introduction substantially increased threat compared to a control condition, ATE = 0.39 *SD* [0.25, 0.53], pr(ATE > 0) > .99, which is consistent with qualitative accounts of the investigatory stop. This increase in threat was highly different from the replicated reduction in threat from the transparency statement, Difference in ATEs = -0.46 *SD* [-0.61, -0.33], pr($ATE_{ambiguous}$ > $ATE_{transparency}$) > .99. Thus, the transparency statement that explicitly conveyed benevolent intent was more effective.

What if an introductory statement communicated benevolent intent, but did so in a way that still left open the possibility that the civilian could perceive it to be an "investigatory stop?" In Experiment 5 ($N = 450$), we used a three-condition design to compare an *ambiguously positive* introduction (i.e., the officer "just wanted to stop and say hi and see how you are doing") to the transparency statement and to a control condition. The ambiguously positive introduction did not meaningfully reduce threat compared to the control condition, ATE = 0.08 *SD* [-0.03, 0.21], pr(ATE < 0) = .18, and that null effect was much smaller than the threat reduction effect obtained from the transparency statement, Difference in ATEs = -0.39 *SD* [-0.53, -0.25], pr($ATE_{ambiguous}$>$ATE_{transparency}$) > .99. A follow-up analysis suggested that this was because participants thought the officer in the ambiguously positive condition was concealing their true intentions (see the SI). Thus, transparency statements seem to reduce threat not simply because they

are friendly, positive, or respectful, but because they more clearly take off the table the possibility that individuals are going to be accused of a crime by law enforcement.

Two final experiments used a different approach to test our assumption that transparency statements are beneficial primarily because they make it clear to the community member that they are not under threat of arrest or bodily harm while they are being questioned. We did this by varying the social role of the interaction partner. In Experiment 6 ($N = 349$), a four-cell study (2 $\times$2 design), as a contrast, participants anticipated interacting with someone who had no authority to arrest or harm them: *a grocery store worker*. In that interaction, there was no reduction in threat in the transparency condition relative to the control condition, ATE = -0.03 *SD* [-0.21, 0.14], pr(ATE < 0) = .59, and that null transparency effect with the grocery store worker was much smaller than an effect obtained from an anticipated police officer interaction, difference in ATEs = -0.38 *SD* [-0.65, -0.11], pr($ATE_{grocery}$>$ATE_{police}$) = .97.

In Experiment 7 ($N = 451$), the final four-cell study, participants anticipated interacting with someone who technically *did* have authority to arrest or harm the participant, but who was very unlikely to do so because it is not common practice in their role: *a park ranger*. When participants anticipated interacting with a park ranger, the transparency statement did reduce threat relative to the control condition, ATE = -0.46 *SD* [-0.62, -0.31], pr(ATE > 0) > .99, but this transparency effect was approximately half the size of the effect obtained when participants anticipated interacting with a police officer, difference in ATEs = -0.37 *SD* [-0.61, -0.16], pr($ATE_{park\ ranger}$>$ATE_{police}$) > .99.

In summary, transparency statements work best—and are most needed—when they shift the default construal of an interaction, from one that could end in harm or arrest, to one that could end in a positive relationship with a powerful authority figure. We note that the goal of these online studies was to establish internal validity with respect to transparency statements and their mechanisms. We do not make claims of generalizability based on these samples.

# Discussion

This research identified a simple and relatively costless method for making community-police interactions less threatening: a transparency statement. In doing so it highlighted the under-appreciated role of *timing* in conversations between institutional authorities (i.e., law enforcement officers) and civilians. Notably, our observations and interviews[22] found that traditional patrol officers typically made transparency statements only after trust had been compromised. Consistent with this qualitative observation, our study showed that absent a transparency statement at the start of the conversation, the officers' questions contributed to a sense of threat that was apparent in an avoidant conversational style (i.e. less authentic, more terse) and physiological profile (lower initial arousal). When officers made transparency statements in the first two sentences of the conversation, however, it changed the tenor of the ensuing interaction. For example, twice as many community members reported feeling *inspired* by the end of the interaction when officers made a transparency statement (28%) than when they did not (14%; see SI and Fig. 1C). This shift was apparent in transcripts, self-reports, and stress physiology.

The present studies raise the intriguing possibility that community policing, in general, might be made more reliably effective if officers could be trained to incorporate transparency statements into their routines. Indeed, in Experiment 1, we individually instructed seven officers in less than five minutes per officer—instruction that might be done efficiently with entire departments. The present study lays the foundation for larger, longitudinal evaluations that assess downstream consequences of transparency trainings on outcomes such as wrongful arrests or official excessive use of force complaints, along with greater trust in the benevolence of police officers, more information sharing, more solved cases, and safer communities.

A key consideration for future research is whether transparency statements could play a role in reducing racial, ethnic, and socioeconomic disparities in policing policies[5,7,39,40]. The results of Experiment 2 suggest that, if anything, people of color[41] and those who have a history of contact with police may profit more from greater transparency[42]. Future experiments, however, should interrogate any potential heterogeneity more systematically, and across multiple levels (individuals and neighborhoods), due to the tight link in the U.S. between individuals' racial, ethnic, and socioeconomic identities and their neighborhood contexts[43].

These findings also have implications for building trust in other organizational relationships with significant power disparities (e.g., border patrol agents, teachers, security guards). We speculate that transparency statements will be especially needed when authority figures can harm those in lower power positions, as is the case for police, principals, teachers and others[38]. In these settings, transparency statements might assure students of their benevolent intent, and might prevent escalating negative reactions from children to these authority figures.

Finally, we note that criminal justice reform efforts often seek to change officer behavior via broad and abstract cultural changes[8,19]. Police departments often resist change, however. Officers remain skeptical of the effectiveness of community policing, or any other reform, as they retain traditionally proactive and aggressive tactics. Our results suggest that reforms that are nearer to officer behaviors—such as interactions with community members—represent an alternative but nevertheless viable pathway to police reform that might be both effective *and* scalable.

# Methods

### Phase 1: Hypothesis Generation via Qualitative Field Observations

In the first phase, we used qualitative methods to understand how community policing efforts might be improved and to generate the hypotheses about transparency statements that we tested in Phases 2-3. Note that we cannot make strong claims from our qualitative methods about fully capturing the dynamics of transparency statements in community policing units. Instead, here we are simply

presenting the systematic approach we took that enabled us to capture ecologically-valid police-community interactions in the field, as recommended by criminology researchers[44]. These field observations ultimately enabled us to develop a falsifiable hypothesis about the potential positive impact of transparency statements in community policing.

**Research settings.** There were two concurrent strands to the field observations we conducted. We began, first, as part of a larger team of researchers observing a new community-oriented policing unit that was developed to address the community's problems by quickly building relationships with community members. In parallel, we observed traditional patrol officers in a different department in the same metropolitan area. We focus here on the most relevant findings about the distinct dyadic interaction tactics law enforcement officers used and their relative effectiveness regarding building trust and rapport with their communities, because these were the observations that led us to develop the present hypotheses. Many other observations were conducted, regarding the governance and decision-making of the two police departments, but these data are less relevant to the present transparency hypotheses.

**Community policing unit.** The first team we observed was a police department in a large, urban city, and the specific unit was located in a district that included portions or entireties of eight neighborhoods with a mixed population of diverse racial and socioeconomic groups (total population range = 14,318 – 93,727; 6-62% White; 15-83% Hispanic/Latinx; 2-78% Black/African-American; 1-6% Asian/Asian-American; 1-3% Other/Multiple Races; Median Age = 32.2 – 41.9 years old; 13-57% population 25+ with a 4-year college degree or greater; 61-94% Native US; 3.5-12.6% unemployed; median household income = $33,515 – $82,908). For example, one beat we observed was 62% White residents with a median income of $72,272. Another beat we observed, in contrast, was only 6% White and 78% Black/African American with a median income of $33,515. This city had a population of more than 2 million residents, and the police department had more than 10,000 officers at the time of our data collection. The unit we focused on was testing a pilot program that reorganized the duties of their officers. While officers traditionally go from call-to-call according to their radios, this unit was given discretion over how they used their time, with the goal to spend it oriented towards building relationships with the community in their assigned area, and coordinating resources that their communities needed. Officers in this unit were recruited from the entire citywide department by the lieutenant in charge of the unit, ostensibly selected on traits that would facilitate relationship-building in communities (e.g., genuine interest in relationship-building, prior history of attempting to build relationships within the constraints of a patrol officer's role). This phase provided us with an understanding of how officers who were ostensibly skilled at building relationships with community members attempted to quickly build trust with civilians. Examples of this process appear in Box 1, right panel, Experiments 1-2.

**Traditional patrol unit.** In parallel, we partnered with another police department in the same metropolitan area to observe more traditional interactions of police whose responsibilities are dictated by radio calls and less focused on developing skills related to connecting with their communities. The employees we observed were patrol officers from a different police department in a moderately-sized, suburban city that was nearby to the city where the community policing unit was located. This city was also relatively racially and socioeconomically diverse (58% White; 12% Hispanic/Latinx; 17% Black/African-American; 9% Asian/Asian-American; 7% Other/Multiple Races; Median Age = 36.2 years old; 67% population 25+ with a 4-year college degree or greater; 82% Native US; 12% in poverty; median household income = $82,335). This city had a population of approximately 75,000 residents, and the police department had approximately 150 officers at the time of our data collection. After discussions with department leadership, we began a four-week long period of four-hour ride-alongs approximately three to four times per week. Per their supervisors' instruction, these officers engaged with members of their community as they naturally would if we were not observing them. We observed these interactions and took observational notes on each interaction we observed, and we also obtained direct reflections from both officers and the community members with whom they interacted. We obtained both structured and unstructured reflections from officers after the interaction with the community member ended. First, officers reflected alone via structured question prompts (e.g., "do you think the civilian understood your intentions for talking with them?"; "how connected to the 'real you' were you in the interaction?"). Then, we discussed officers' experiences utilizing unstructured questions. Simultaneously, while the officer left the area to reflect alone, we approached the community members they talked to for an optional, consented debriefing discussion of their interaction utilizing a mix of structured and unstructured questions. This phase provided us with an understanding of how officers who were not trained in the community-based policing attempted to quickly build trust with civilians. An example of this process appears in Box 1, left panel, Ex 2.

**Participants.** Participants for both phases of the qualitative field observations were the police officers in the two departments we observed (70% Male; 47% White; 30% Hispanic/Latinx; 20% Black/African-American; 3% Asian/Asian-American; approximate age range: 20-60 years old). In total, more than 35 police were observed across three years, which included interviews, ride-alongs, observed unit meetings, and field observations of community events, resulting in more than 500 hours of data observations. In the community policing unit, we observed 17 officers, four sergeants, one lieutenant, and one commander. Throughout the three years of our observation, more than five officers left the unit due to injuries or to pursue other opportunities in the department. In the traditional patrol unit, we observed 18 patrol officers and approximately 148 civilian interactions.

**Data and qualitative analysis.** Data from both phases come largely from field observations of ride-alongs with the police participants. For the ride-alongs, police were asked to act as they naturally would while we observed from a distance. Field observations of community events, biweekly debriefing meetings, and hour-long interviews were only collected from the community policing unit. Biweekly department meetings in this phase typically consisted of strategic discussions about how to approach community problems together in a roundtable with the whole unit. Interviews occurred in four rounds

and consisted primarily of officer views of their role and their community while in the pilot program. Interviews occurred at the recruiting, beginning, middle, and end stages of the officers' involvement in the pilot program. As mentioned, for the traditional patrol unit, observations of the ride-alongs were accompanied by both officers' structured and unstructured reflections alone and with the researcher(s) accompanying them on the ride-along. For all field notes, shorthand notes were recorded in small notebooks or on smartphones and expanded after the conclusion of the observation, as soon as possible. Recorded conversations were transcribed by a professional transcription service.

Field notes and interview transcripts were analyzed by the two lead researchers by discussing the day's data collection immediately or soon after it finished, and writing informal, conceptual, theoretical memos to summarize the emerging themes. Transcripts and notes that were flagged as indicative of key themes were then analyzed line-by-line to understand recurring patterns and to check assumptions against the data. Once the key themes were identified and refined (presented in Box 1), then exemplary quotations were selected to illustrate those themes to readers. These quotations were then vetted with a third researcher, to reach agreement on the fit between the data and our interpretations of the data. After reducing the qualitative data to the key themes and focal examples (Box 1), we used these insights to design the randomized experiments and test the hypotheses directly. Additional thematic examples appear in the SI.

## Phase 2: Hypothesis Testing in Behavioral Experiment 1

**Participants**. The study was pre-registered on the Open Science Framework (OSF; tiny.cc/PreRegExp1). We planned to collect data from a total of 200 participants, which afforded the ability to detect an effect of 0.4 *SD* at 80 percent power using conventional null hypothesis rejection analyses. Following the pre-registered stopping rule, data collection was terminated on the first day that we met or exceeded our sample size. No data were examined and no hypotheses were tested prior to stopping data collection.

We ultimately collected data from 239 U.S. residents from a sample of community members in a large urban city from streets nearby, Craig's list ads, emails throughout departments to the local university, and word of mouth. After excluding seven individuals who either opted to have their data deleted after learning the full details of the study ($n$ = 3) or had incomplete data ($n$ = 4), we were left with a final sample of 232 ($M_{age}$ = 22.07, $SD$ = 5.90; 54 percent female, 42% male, 4% non-binary/genderqueer/other; 34% Whites/European Americans, 32% Asian/Asian Americans, 17% Latino/a/Hispanic Americans, 9% Black/African Americans, 5% Multiethnic, 3% Biracial; conservatism $M$ = 2.68, $SD$ = 1.35).

**Material development**. The study design was created with informal police consultants, who were current and retired police officers from three departments. We developed our procedure to realistically reflect

what police do in officer-initiated interactions with their communities, while maintaining alignment with our definition of transparency (i.e. clarifying intentions for approaching civilians).

**Civilian participants**. This study was conducted during June and July of 2021. Civilian (i.e., community member) participants were told that the study was about the natural interactions people had with others. Civilian participants were seated individually at a table around a university building in one of five predetermined locations that were not in-view of each other (e.g., a patio table; a bench). Civilian participants were told that they should do what they would naturally do if they were sitting by themselves outside (e.g., using their cell phones or computers). The research assistant left the participants alone with an iPad and stated that everything would be audio-recorded by the iPad during the experiment. Participants did not know who they would be interacting with but were told to act naturally if (and when) anyone approached them.

To mask the anticipation of a police officer approaching while also allowing for informed consent, participants were told in their consent form that due to the outdoor nature of the research study, they may be approached by a research assistant, unsheltered (i.e., homeless) person, student, police officer, or others in the area. Once the conversation started, the officers never stated that they were a part of the research study.

**Police officer partners**. The police officer partners ($N = 7$) for this field experiment were employees of the police department with jurisdiction over the community of recruited civilian participants. That is, actors were not used, and all officers in the study were on-duty employees during their regularly scheduled work shifts, not trained confederates; this decision was made to increase the ecological validity of the experiment.

Prior to conducting this research, the research team met with police department leadership, who approved the partnership. The seven police officers were recruited by the leadership because they were considered by leaders to be representative of the diversity of officers using community policing efforts in this police department. Training of the officers lasted approximately five minutes per officer. The officers overall spent approximately 60 hours on data collection in the role of an experimenter.

Police partners were told to initiate interactions with civilians in one of two ways: either providing a transparency statement (i.e., giving explicit reasons for why they are approaching the participant, at the outset of the interaction) or not (our control condition; i.e., using the ambiguous, direct, and short language that simply asks if they can talk to the civilian participants). Police partners were instructed to keep their conversations short (i.e., to five minutes or less, aiming for 2 minutes), but were otherwise left to their own natural conversation style after the introduction. They did not follow an exact script. Notably

transcripts confirmed that 100% of the officers in the transparency condition made a transparency statement, and 0% of the officers in the control condition did so at the outset of the conversation.

For example, in the transparency condition ($N$ = 100), officers initiated interactions in the following ways: e.g., “Hello, I’m just out and about walking around, talking to people in the community. Is it okay if I talk to you for a minute?” “Hey, my name is [Name], and I’m just taking a walk, trying to get to know my community better. You mind if I sit and talk to you for a second?”

In contrast, in the control condition ($N$ = 132), officers initiated interactions in the following ways: e.g., “Hey, can I talk to you for a minute?” “Hey, I’m Officer [Name], can I talk to you for a second?”

**Assignment to condition**. Leadership in the police department assigned officers to one of several shifts for the experiment. Researchers, blind to the identities or skills of the officers, assigned officers’ shifts to experimental conditions, so that in some shifts officers used transparency statements, and during other shifts they did not. Civilians, in turn, were assigned to different shifts independently of the officers’ condition assignments. There were an equal number of weekend shifts across conditions. Within experimental conditions, shift was not meaningfully associated with the pre-registered outcomes of threat and trust (Threat: Control group intraclass correlation [ICC] = .04; Transparency group ICC = .02; Trust: Control group ICC = 0; Transparency group ICC = 0). As a result, supplementary, non-pre-registered mixed-effects analyses that clustered standard errors by shift did not produce different results (Threat: $b$ = -0.47, $t$ = -2.09, $p$ = .037; Trust: $b$ = 0.37, $t$ = 2.23, $p$ = .025). Furthermore, no participant demographics (i.e., age, gender, race, native U.S. status, education level) differed significantly by condition, $t$’s ≤ 1.106, $p$’s ≥ .27 (see Table S2 in SI) or weekday (see Table S3 in SI).

## Measures

Unless otherwise acknowledged, all measures were on a seven-point Likert scale, ranging from 1=*Strongly Disagree* to 7=*Strongly Agree*. For all measures, civilian participants read that their individual data would not be shared with the police (i.e., that data collection was confidential). Full lists of the scale items and additional information can be found in the SI. See Table S4 in the SI for correlations among all primary variables.

**Manipulation check: Perceptions of transparency of intentions**. As a manipulation check, we measured civilian participants’ perceived clarity of the police partners’ intentions (i.e., “The officer I talked to stated their intentions for talking to me clearly”; “The officer I talked to stated their intentions for talking to me immediately”; “I knew why the officer was talking to me from start to finish”; α = .757).

**Primary measure 1: Threat of enforcement**. Civilians’ self-reports of threat were measured using three items that operationalized threat as the expectation of harm in the interaction—i.e., the inverse of trust. Specifically, these questions asked how likely (1=*Very unlikely* to 7=*Very likely*) it was that the police officer they talked to “just wanted to get to know you,” (reversed) “just wanted to make friendly

conversation," (reversed) or "did not trust you" α = .768. Moreover, a principal components analysis of these three items indicated that all three items loaded highly (loadings ≥ 0.81) onto a single component accounting for 77% of the total variance.

**Primary measure 2: Trust in benevolent intent**. Trust is viewed as the first step towards repairing relationships between law enforcement and civilians in the community policing model[14]. Civilian participants' trust was measured using three items adapted from the benevolence subscale of a validated trust scale[45], using the same likelihood scale as threat, regarding future interactions with the police officer they were approached by: "This police officer would care about you and your welfare;" "this police officer would go out of their way to help you; this police officer would not do anything to hurt you." α = .630).

### Natural Language Analysis

**Audio recordings of conversations.** Audio was recorded by an iPad device positioned on the table where the civilian participant was seated. Audio began recording when the research assistant set the participant up alone during the pre-interaction waiting period through when the interaction ended, when the research assistant returned to turn the recording device off. All audio recordings were later transcribed and turned into text files by research assistants.

**Natural language processing: Conversation transcripts.** Text analyses of transcripts of the conversations between civilians and officers were divided by speaker role (i.e., officer vs. civilian), condition (i.e., transparency vs. control), and by portion of the conversation (i.e., first, middle, or last third). Transcripts were analyzed with the Linguistic Inquiry and Word Count (LIWC) Software[29].

LIWC is one of the most dominant text analysis software in the social sciences, and relies on a "word counting" method that both tallies counts of parts of speech like prepositions, adverbs, and pronouns[46], and also includes four summary dimensions validated in previous research: Analytic[47], Clout[48], Authenticity[49], and Emotional Tone[50]. We analyzed the end of the civilian side conversation transcripts and obtained their score on each of these summary dimensions. The Authenticity summary dimension included in LIWC is a composite score validated in prior research and reflects honest, unfiltered, and spontaneous speech[49]. It is a bottom-up composite measure that was built based on research on deception and language use. The language categories that comprise the authenticity dimension resulted from factor-analyzing texts of truthful vs. deceptive language[51]. We expected that the transparency statement would increase civilians' use of authentic language, because the transparency statement puts civilians at ease and enables rapport to be built. We did not anticipate differences on any of the other dimensions, but analyzed them along all summary dimensions to document the specificity of the effect of transparency statements on authentic language.

**Natural language analysis: Civilian reflections on the interaction.** On the post-interaction survey, participants answered an open-ended question asking about the thoughts (one question) and feelings (a

second question) they experienced at the end of the interaction. These reflections were also scored with the LIWC-22 algorithm. Analyses focused on two patterns that emerged from the idiographic analysis. The first is *causal* language, or signaling people's tendency to ponder the cause of the officer's interactions (i.e. "I was wondering why he was talking to me"). We expected the transparency statement to reduce causal language in the reflections, because the primary goal if the transparency statement is to answer the question of why the officer is talking to them. The second was *positive emotion* (e.g. comfortable, happy). We expected the transparency statement to increase positive emotion at the end of the interaction, because the statement should provide the foundation for rapport that could result in a positive back-and-forth dialogue.

**Emotion at the end of the interaction.** On the post-interaction survey, civilian participants completed the Short Form version of the Positive and Negative Affect Schedule (PANAS-SF[52]), selecting an emotion they felt at the end of the interaction. In the standard PANAS-SF measure, there are several emotion categories that correspond to operationalizations of threat (vs. challenge) appraisal[31–33]. Emotions were coded for *threat* (1 = alert, afraid, nervous, upset, hostile, or ashamed; 0 = not). In addition, one emotion (*inspired*) was explored, due to its alignment with community policing's goals.

**Skin conductance levels (SCL).** To distinguish between challenge (approach) and threat (avoidance) stress levels, skin conductance levels (SCL), were obtained from electrodermal activity (EDA) sensors embedded in a wrist-worn ambulatory device: the Empatica E4. The Empatica E4 wristband measures physiological responses to stress simultaneously via two sensors placed on the bottom of the wrist[53,54]. The E4 sensors have a resolution of 1 digit ~900 picoSiemens, a range from 0.01microSiemens to 100 microSiemens and they apply alternating current (8Hz frequency) with a maximum peak to peak value of 100microAmps (at 100microSiemens).

EDA assesses changes in the amount of conductance of the skin (i.e., SCL) due to sweat produced by *eccrine* glands; increased SCLs correspond to increased sympathetic nervous system (SNS) activation innervated by acetylcholine[33].

We instructed participants to push a button our ambulatory devices whenever someone started to talk to the participant. Pushing this button recorded a timestamp in the ambulatory device data. Because of this participant-led action, some participants forgot to push the button to indicate the interaction started. This is a potential limitation because it caused missing psychophysiological data (the SCL subsample was $N = 177$). However, this missing data was unavoidable to retain the realism of the unexpected officer-initiated interactions.

**SCL theoretical predictions.** According to the BPS model, SNS arousal during a stressful interaction can follow one of two patterns: a more positive, *challenge-type* response or a more negative, *threat-type* response. A challenge-type response corresponds to approach motivation and evokes an initial orienting response with the onset of the stressor (e.g. the approach of the officer), which involves a rapid increase in SNS activity[30–32]. But a challenge-type response would involve a rapid return to homeostasis as the

interaction is experienced as not a potential threat[33–35]. In contrast, more negative, avoidance-oriented, threat-type responses often involve blunted initial SNS reactivity, as individuals seek to avoid potentially harmful threats (as is evident from the transcript of the control group participant in Fig. S2A in SI). When individuals cannot escape the stressor, though, those experiencing threat-type responses remain suspicious, vigilant, and alert to impending harm, thus leading their SNS activation to increase over time[33–35].

As shown in the SI, we validated these expectations with respect to the specific SCL facets extracted from the ambulatory EDA data by using a pilot laboratory study ($N$ = 100) that involved an acutely stressful interaction (the Trier Social Stress Test; TSST[55]) and gold-standard measures of challenge and threat-type responses (Total Peripheral Resistance, TPR[30]. In the pilot study shown in Fig. S2A-B in the SI, participants showing challenge-type responses (low TPR) showed an initial increase in SCL at the start of the TSST, as expected, and a sharper decline during recovery, compared to those with threat-type responses (high TPR).

**EDA processing.** To calculate skin conductance levels (SCLs), the EDA data were extracted from the devices after each session and processed in Python using established algorithms for processing EDA data and removing artifacts. To check EDA quality, the algorithm first marked the lost signal and calculated the ratio of lost versus overall signal. The signal was considered lost if the value was below 0.001μS. If the ratio was greater than 0.9 for each five-second window, the algorithm classified the window as of bad quality[56]. The algorithm then checked any anomalies in EDA data in each second window. The data were considered anomalies if the maximum increase of a signal value was greater than 20% and the maximum decrease was greater than 10%[57]. The segment was marked as of bad quality if this condition was not met in each second window. Finally, the algorithm calculated signal power in second difference from the SC signal in a window of five minutes with four minutes overlap[58]. The algorithm set the four-minute overlap to achieve a resolution of smoothed processed data per minute[59].

**Second derivative of EDA.** The SCL metric of interest was calculated as the average second derivative of EDA trends over 10-second intervals. When the skin conductance level is moving toward a “spike” then that is an indication of greater engagement with a stressor, and that is therefore the primary physiological response that was relevant to the BPS model predictions. The way to assess whether, in a particular moment, a person is approaching a spike is to assess the second derivative of the time function. The first derivative, of course, is equal to the slope—whether SCL is increasing, and to what extent—but this does not indicate whether a spike is approaching because it does not reveal anything about the rate of change in the slope, and therefore it cannot indicate peaks or troughs in the data. When the second derivative is positive, it means that the slope is increasing—for example, that it is the concave part of the time function. When the second derivative is negative, it means that the slope is decreasing—i.e., that it is in the convex part of the time function. Note that a negative second derivative does not mean that SCL is no longer increasing—only that the rate of change is slowing. Likewise, a positive

second derivative means that the slope for SCL is increasing—even if it is still in the "downhill" part of the time function. The primary advantage of the second derivative is that it can be a leading indicator of a coming spike in the SCL, and therefore it can make fuller use of the data (as compared to waiting for the spike to appear in the data). For example, a 20-second interval of data sampled might have one spike but 20 second derivatives.

**Exclusion of PPG data.** The E4 wristband also has photoplethysmography (PPG) sensors, but the PPG data were not used in the present paper. The PPG sensor shines light on the skin to detect the degree to which light refracts. Fluctuations in blood flow are estimated based on assumptions about blood flow, using the default Empatica software algorithms. Heart rate (HR) and interbeat intervals (IBI) are received as output along a time-series, with estimations at approximately 64 Hz. However, when validating the measurement of the PPG sensors using the gold standard for cardiovascular activity (i.e., electrocardiogram: ECG), our pilot study showed that the E4's IBI algorithm was only able to extract 33.7% through the TSST (Trier Social Stress Test[39]) and 17.8% in readings from a controlled lab setting in schools. To our knowledge, heart rate measurement of E4s have not been validated in naturalistic settings like ours in a published paper. In naturalistic, outdoor settings like ours in Experiment 1, there is likely light leak and movement artifacts that created error in measurement that accounts for differences in measurement accuracy recorded in traditional lab settings that have had better success with PPG sensors[53,54]. Data from our pilot experiment left us less confident in any measurements of heart rate. However, data from our pilot experiment (see below) gave us confidence in using E4 EDA measurements of skin conductance.

### Phase 3: Replication and Exploration of Moderation by Demographic Group Membership

### Experiment 2

Participants first read a scenario where an officer walked up to them while they were outside in a park with their friends. Then, participants were exposed to a brief video recording of an officer initiating an interaction before completing our focal measures of threat, trust, and behavioral intentions towards the officer and other officers in the future.

**Participants.** We preregistered our experiment on OSF (http://tiny.cc/VideoPreReg). We aimed for a sample size of approximately 1000 and recruited a sample of 969 U.S. residents from Prolific that was representative of the U.S. population in terms of age, gender, and race/ethnicity. We made no exclusions ($M_{age}$ = 45.67, *SD* = 16.44; 52% female; 76% Whites/European Americans; 6% Asian/Asian Americans; 4% Latino/Hispanic Americans; 12% Black/African Americans; 0.02% Native American, 0.02% Hawaiian/Pacific Islander, 2% Bi- or Multi-racial; $M_{income}$ = 6.43, *SD* = 3.39; 53% 4-year college degree or greater; economic conservatism *M* = 3.51, *SD* = 1.88; social conservatism *M* = 3.13, *SD* = 1.83). Analyses were powered to detect an effect of 0.10 *SD* at 80% power.

**Design and procedure.** Participants were randomly assigned to one of two conditions: control or transparency. For participants in the control condition, in the video, the police officer simply started a

conversation by saying “Hi, how are you doing?” In the transparency condition, the officer explained that they were trying to get to know the community better (see below).

1. You walk into a park with one of your friends.

2. You sit down in the grass and talk with your friend for about 30 minutes.

3. You notice a police officer park their car nearby and see them get out of their car.

4. The police officer walks around for about 10 minutes, eventually looks your way, and walks up to you.

“Hey there, I'm [Officer Name], I'm just out here saying hi and seeing how things are going.” [Control Condition]

“Hey there, I'm [Officer Name], I'm out here on foot patrol, just trying to get to know my beat and my community.” [Transparency Condition]

**Threat and trust.** Participants answered an expanded 10-item measure about perceived threat that included the three focal items from Experiment 1 and added 7 additional items to more broadly encapsulate the experience of threat in the interaction (e.g., How likely is it that… “The police officer is going to accuse you of something.”, “The police officer does not trust you.”; α = .96) and perceptions of trust from the integrity subscale of a trustworthiness scale (i.e., How likely is it that this officer… “...has a strong sense of justice. “...tries hard to be fair in dealing with others.” α = .92). Importantly, this expanded measure included the same 3-item composite as in all other studies. The three-item composite was also reliable (α = .85).

**Behavioral intentions.** Participants also answered 6 items about their willingness to initiate an interaction with the focal officer and other officers from the same department in the future (e.g., In the future, how likely would you be to do the following?… “Say hello when you see this officer [another officer from this organization]”, “Wave if you see this officer [another officer from this organization]”, “Start a conversation with this officer [another officer from this organization]”; $\alpha_{focal}$ = .92; $\alpha_{other}$ = .93).

### Phase 4: Hypothesis Confirmation with Online Vignette Replication Experiments

### Experiment 3

Participants read a vignette about a police officer stopping to talk to them after they left a grocery store. A grocery store was chosen because it mirrors the kind of mundane situation in which civilians are approached by officers engaging in community policing. Also, a grocery store is a place where an officer could plausibly be investigating a crime, and so it was a situation in which the officer’s intent could be ambiguous to the participant.

**Participants.** We preregistered our experiment on OSF (http://tiny.cc/5goyuz). We aimed for a sample size of approximately 300 per condition and recruited a sample of 649 U.S. residents from Amazon’s

MTurk, of these 609 completed our dependent measures. We made no exclusions ($M_{age}$ = 37.00, *SD* = 12.28; 52% female, 1% non-binary/other; 72% Whites/European Americans; 8% Asian/Asian Americans; 6% Latino/Hispanic Americans; 11% Black/African Americans; 1% Native American, 1% Other). Analyses were powered to detect an effect of 0.29 *SD* at 80% power.

**Design and procedure.** Participants were randomly assigned to one of two conditions: control or transparency. Participants were asked to imagine walking in and out of a grocery store. For participants in the control condition, the police officer simply started a conversation by saying "Hi, how are you doing?" where the intent was ambiguous. In the neutral transparency condition, the officer explained that they were trying to get to know the community better (see below).

1. You leave the store.

2. On your way out, you notice a police officer's car pull next to where you were walking.

3. The police officer walks out of the car and looks your way.

4. The police officer walks towards you and starts a conversation, saying,

"Hi, how are you doing?" [Control Condition]

"Hi, how are you doing? My boss is asking me to talk to people to get to know the community better." [Neutral Transparency Condition]

**Threat and trust.** Participants were assigned to either answer items about perceived threat (e.g., How likely is it that… "The police officer is going to accuse you of something.", "The police officer does not trust you."; α = .85) or perceptions of trust from the benevolence subscale of a trustworthiness scale (i.e., How likely is it that this officer… "...is very concerned about my welfare." "...would not knowingly do anything to hurt me." "...will go out of his/her way to help me." "...has a strong sense of justice. "...tries hard to be fair in dealing with others." [1] α = .84). See Table S10A-B in the SI for correlations among all primary variables.

## Experiment 4

**Participants.** We preregistered our study on OSF (http://tiny.cc/6goyuz). Given the effect size obtained in Experiment 3 (*d* = 0.50), we sought to obtain a sample large enough to detect this size effect, but across three conditions. Thus, we recruited a sample of 382 U.S. residents from Amazon's MTurk. We made no exclusions ($M_{age}$ = 38.52, *SD* = 12.54; 51% female, 0.3% non-binary/other; 77% Whites/European Americans; 11% Asian/Asian Americans; 8% Latino/Hispanic Americans; 10% Black/African Americans; 2% Native American, 1% Hawaiian/Pacific Islander, 2% Other). Analyses were powered to detect an effect of 0.32 *SD* at 80% power.

**Design and procedure.** Participants were randomly assigned to one of three conditions: control, community-oriented transparency, or aggressive policing. All participants read a vignette about a police officer stopping to talk to them after they left a grocery store. The control and community-oriented transparency were identical to the two conditions of Experiment 3, while in the aggressive policing condition the officer explained that they were trying to find a suspect (see below). After reading this vignette, participants answered four items about the amount of threat they perceived from the police officer as in Experiment 3.

1. You leave the store.

2. On your way out, you notice a police officer's car pull next to where you were walking.

3. The police officer walks out of the car and looks your way.

4. The police officer walks towards you and starts a conversation, saying,

"Hi, how are you doing?" [Control Condition]

"Hi, how are you doing? My boss is asking me to talk to people to get to know the community better." [Neutral Transparency Condition]

"Hi, how are you doing? "Hi, how are you doing? My boss is asking me to get to know the community better to find a suspect we're looking for." [Ambivalent Transparency Condition]

**Threat.** We measured threat using the same scale as in Experiment 3. See Table S11 in the SI for correlations among all primary variables.

### Experiment 5

**Participants.** We preregistered our study on OSF (http://tiny.cc/7goyuz). We sought to recruit a similar size sample to that of Experiment 4, so we recruited a sample of 450 U.S. residents from Prolific Academic. We made no exclusions ($M_{age}$ = 31.59, *SD* = 11.05; 50% female, 2% non-binary/genderqueer/other; 60% Whites/European Americans; 17% Asian/Asian Americans; 8% Latino/Hispanic Americans; 10% Black/African Americans; 0.2% Native American, 2% Biracial, 3% Multiethnic). Analyses were powered to detect an effect of 0.29 *SD* at 80% power.

**Design and procedure.** In Experiment 5, we asked participants to read a scenario where an officer walked up to them while they were wearing masks outside in a park with their friends.[1] Participants were randomly assigned to one of three conditions: control, transparent, or ambiguously positive. The officer either asked, "Hi, how are you doing?" (control condition), gave a transparent and concrete reason ("I'm trying to get out of my car every day to talk to people and get to know the community better"), or an ambiguously positive reason ("Just wanted to stop and say hi and see how you are doing").

Imagine the following happened to you...

1. You walk into a park with one of your friends. You are both wearing your masks.

2. You sit down in the grass and talk with your friend for about 30 minutes.

3. You notice a police officer park their car nearby and see them get out of their car.

4. The police officer walks around for about 10 minutes, eventually looks your way, and walks up to you.

5. The police officer starts a conversation, saying,

"Hi, how are you doing?" [Control Condition]

"Hi, how are you doing? Just wanted to stop and say hi and see how you're doing." [Ambiguously Positive Condition]

"Hi, how are you doing? I'm trying to get out of my car every day to talk to people and get to know the community better." [Concrete Transparency Condition]

**Threat.** We measured threat using the same scale as in Experiments 3-4.

**Perceived authenticity.** In addition to the threat items measured in previous experiments, we added a measure of perceived authenticity ("How [ 'authentic,' 'sincere,' 'fake' (reversed)] do you think this office is being with you?"; α = .935).

**Additional demographics.** Additional control variables for education, political ideology, social connection to police, and being a US native. In our sample, 39% of participants reported knowing at least one person who was or had previously been employed in the field of law enforcement. Our sample included 20% Republicans, 47% Democrats, 24% Independents, 3% Other, and 7% No Preference (economic conservatism: $M$ = 3.41, $SD$ = 1.65; social conservatism: $M$ = 3.09, $SD$ = 1.71). See Table S12 in the SI for correlations among all primary variables.

### Experiment 6

**Participants.** We preregistered our study on OSF (http://tiny.cc/8goyuz). We sought to recruit a similar size sample to that of Experiment 4, so we recruited a sample of 349 U.S. residents from Amazon's MTurk. We made no exclusions ($M_{age}$ = 37.55, $SD$ = 11.36; 53% female, 0.3% non-binary/other; 76% Whites/European Americans; 9% Asian/Asian Americans; 7% Latino/Hispanic Americans; 12% Black/African Americans; 2% Native American, 1% Hawaiian/Pacific Islander, 0.3% Other). Analyses were powered to detect an effect of 0.33 $SD$ at 80% power.

**Design and procedure.** The design and procedure were identical to that of Experiment 4, except that we manipulated target and removed the aggressive policing condition used in Experiment 4. This resulted in a fully-crossed 2 (Target: Police, Grocery Store Worker) × 2 (Transparency: Control, Transparent) design.

Imagine the following happened to you...

1. On your way to the grocery store, you see another person walk in before you.

2. You walk behind them into the grocery store and head in to start your shopping.

3. You leave the store.

4. On your way out, you notice a [police officer's car / grocery store worker who is moving carts] pull next to where you were walking.

5. The [police officer walks out of the car / grocery store worker parks the carts] and looks your way.

6. The [police officer / grocery store worker] walks towards you and starts a conversation, saying,

"Hi, how are you doing? My boss is asking me to talk to people to get to know the community better." [Transparent]

"Hi, how are you doing?" [Control]

**Threat.** We measured threat using the same scale as in Experiments 3-5. See Table S13 in the SI for correlations among all primary variables.

### Experiment 7

**Participants.** We preregistered our study on OSF (http://tiny.cc/agoyuz). We sought to recruit a similar size sample to that of Experiment 6, so we recruited a sample of 451 U.S. residents from Amazon's MTurk. We made no exclusions ($M_{age}$ = 37.86, *SD* = 12.87; 58% female, 0.2% non-binary/other; 71% Whites/European Americans; 8% Asian/Asian Americans; 5% Latino/Hispanic Americans; 8% Black/African Americans; 1% Native American, 5% Bi- or Multi-racial, 1% Other). Analyses were powered to detect an effect of 0.29 *SD* at 80% power.

**Design and procedure.** Participants read a similar vignette to Experiment 6 that instead took place at a park. The study again comprised a fully-crossed 2 (Target: Police, Park Ranger) × 2 (Transparency: Control, Transparent) design. Participants read that a transparent (or control) police officer (or park ranger) walked up to them.

1. You head to a nearby park.

2. While at the park, you notice a [police officer / park ranger]'s car pull next to where you were walking.

2. The [police officer / park ranger] walks out of the car and looks your way.

3. The [police officer / park ranger] walks towards you and starts a conversation, saying,

"Hi, how are you doing?" [Control]

“Hi, how are you doing? My boss is asking me to talk to people to get to know the community better.” [Transparent]

**Threat.** We measured threat using the same scale as in Experiments 3-6. See Table S14 in the SI for correlations among all primary variables.

### Data Availability

All data supporting the findings in this manuscript are available on the Open Science Framework and can be found at the following link: http://tiny.cc/TransparencyProject

### Code Availability

All code for analyses supporting the findings in this manuscript are available on the Open Science Framework and can be found at the following link: http://tiny.cc/TransparencyProject

[1] In Experiment 5, we replicated our effect in the midst of social distancing mandates during the pandemic and unrest from protests against the police after the murders of George Floyd and Breonna Taylor, in 2020.

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# Declarations

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*Author contributions.* The conceptualization of the research hypotheses and the design of the study were initiated and led equally by K. Dobson and A. Dittmann. Dobson and A. Dittmann supervised all data collection and processing by research assistants, conducted all field observations and interviews, produced the datasets, scored transcripts using natural language processing transcripts, and drafted the supplemental materials. D. Yeager conducted the Bayesian analyses presented in the main text and edited the supplemental materials. All authors wrote, edited, and approved the final manuscript and supplemental materials.

*Competing interests.* The authors have no competing interests to declare.

*Data and materials availability.* Preregistration details, code for analyses, anonymized data, and experiment materials can be found at http://tiny.cc/TransparencyProject

# Figures

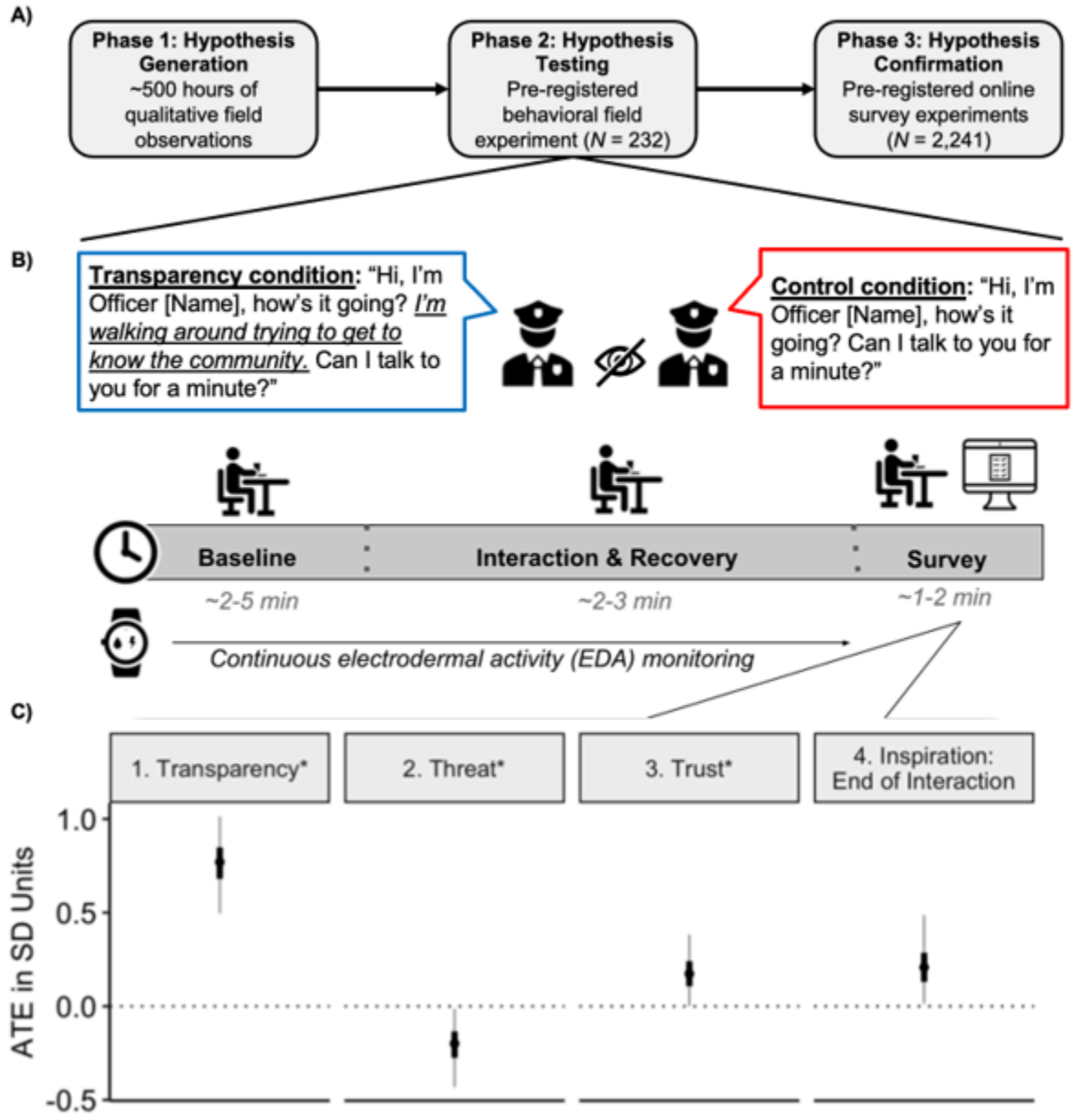
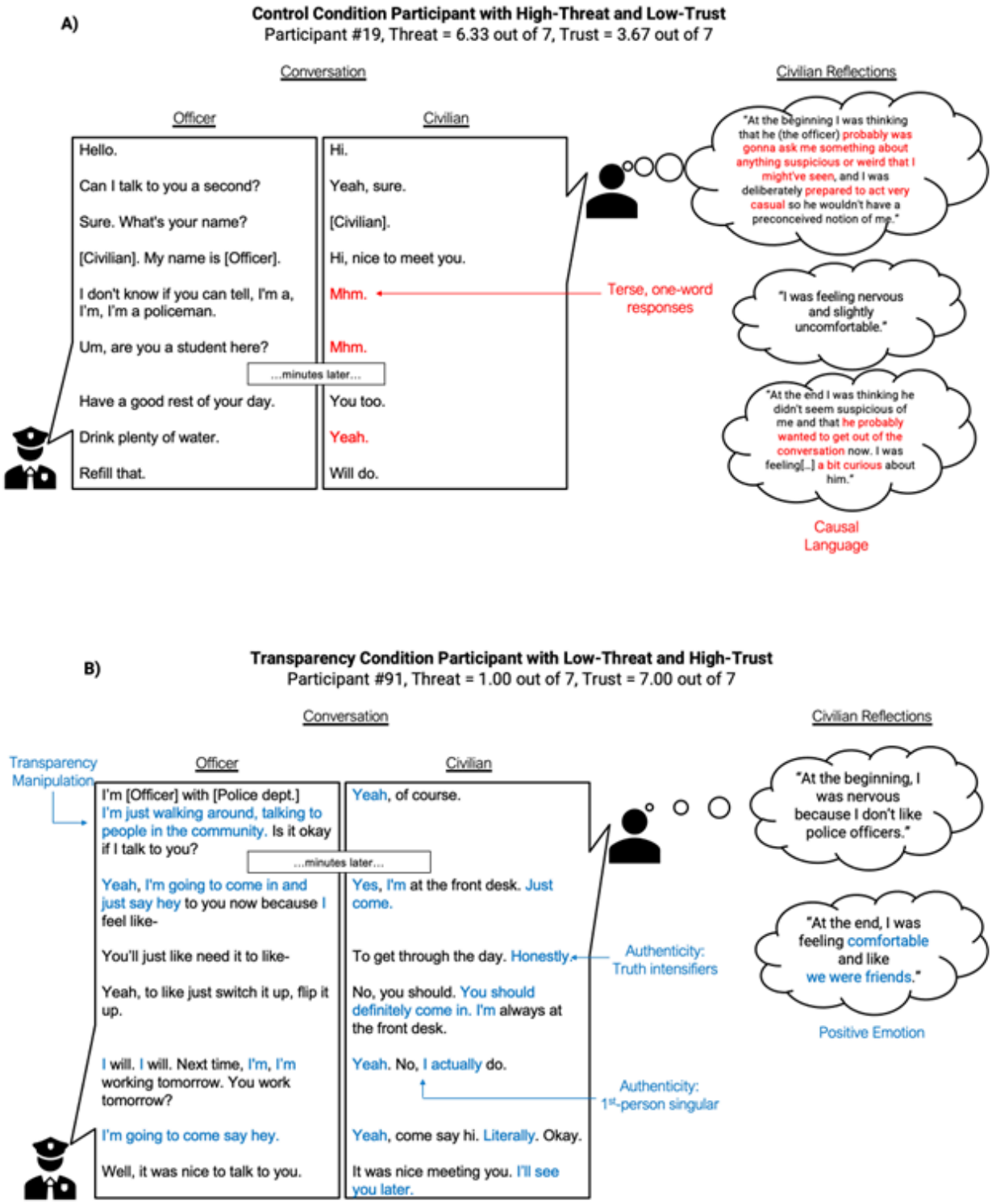



**Figure 1**

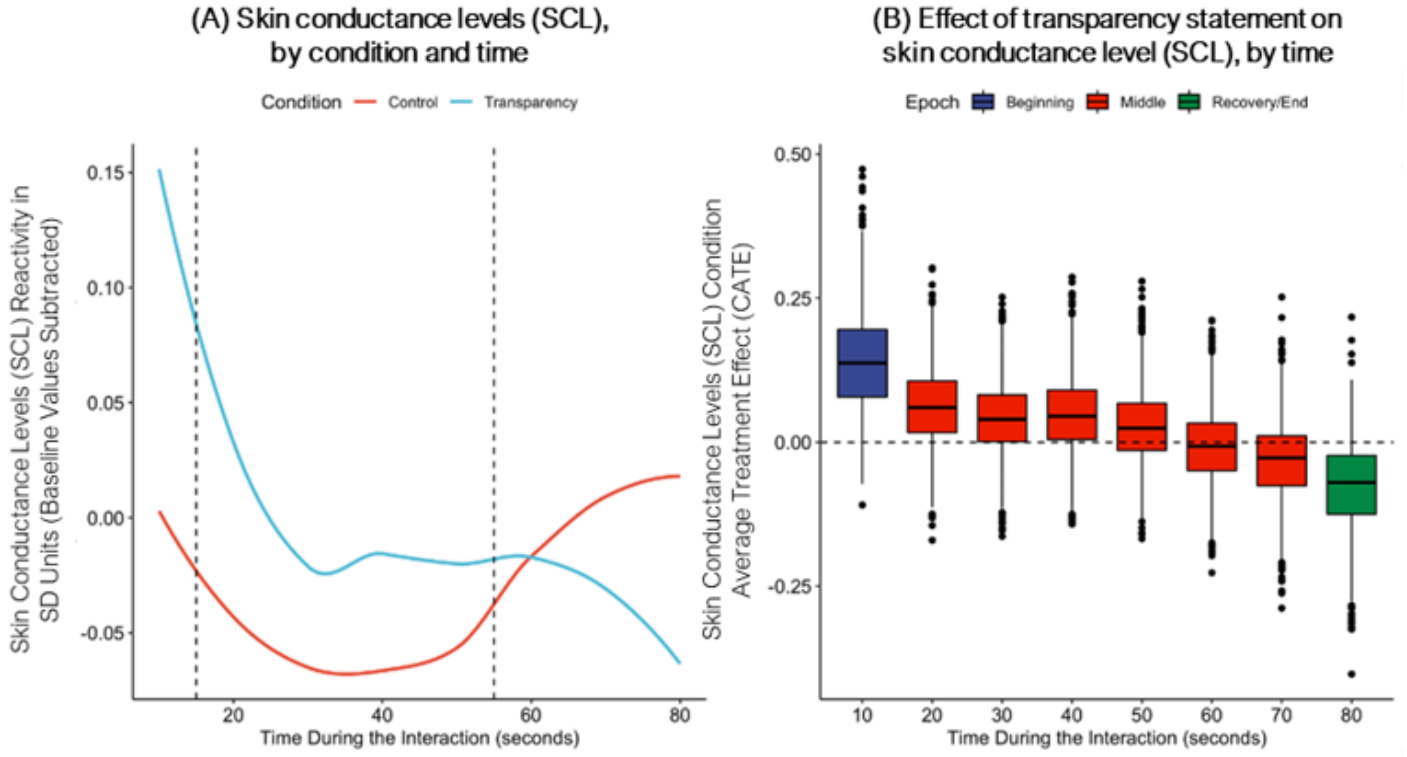
**A) The three phases of the present research, B) The behavioral study procedures, and C) average treatment effects (ATEs) of the transparency statement in the behavioral study.** Note: * = pre-registered, confirmatory outcome. ATE = average treatment effect. EDA = electrodermal activity, which assessed sympathetic nervous system (SNS) activation (see Fig. 3). B) The non-seeing eye represents assignment of community member participants to experimental condition, with the participants and experimenters both kept unaware of condition assignment. Officers were aware of condition assignment because they

delivered the manipulation verbally, but study hypotheses were not shared with them. Participants recruited for the study were seated with the sensors during a baseline period, told to act as they naturally would while sitting outside by themselves. Participants did not know when or who they would be approached by, but were told the sensors would monitor them during any interactions they had while in the study. C) ATEs calculated in a Bayesian Causal Forest model that used a machine-learning approach with conservative BART priors; covariates were age, race/ethnicity, gender, education level, and English language proficiency of community members. Thick bars represent the inter-quartile range (IQR) of the posterior distribution of treatment effects, and thin bars depict the 10th to the 90th percentiles. Statistical tests for the final outcome—inspiration—appear in the SI. Conventional null hypothesis tests were all $P < .05$ (see the SI).




## Figure 2

**Idiographic analysis of transcripts from police-community interactions (left) and post-interaction reflections (right) for participants in (A) the control condition and (B) transparency condition.** Note: Phrases are highlighted to illustrate examples of differences in natural language scored via the LIWC-22 algorithm that were meaningfully different across conditions in statistical analyses of all participants' data (see the main text).




## Figure 3

**Differences in the temporal dynamics of the police interaction across experimental conditions depicted as (A) overall skin conductance level (SCL) and (B) conditional average treatment effects on SCL.** Note: SCL = Skin conductance level. Results in panels A and B were generated by a multilevel random intercept model that used Bayesian Additive Regression Trees (BART), a machine-learning algorithm with conservative prior distributions, to estimate the non-parametric effect of time on SCL and on the treatment effects. Panel A presents loess-smoothed means of the posterior distribution of SCL reactivity estimated by the BART model. In panel B, dark lines correspond to the median of the posterior distribution of effects, boxes correspond to the inter-quartile range, and whiskers correspond to the 2.5$^{th}$ to 97.5$^{th}$ intervals. SCL was indexed by the average second derivative of EDA with respect to time during each 10-second window. A positive second derivative signals more rapid increases in EDA, which signals an impending peak and thus an SCL event, while a negative second derivative signals that an SCL event is ending and therefore the participant's SNS arousal was in the process of recovering to homeostasis. The second derivative was used to follow recommendations from papers validating SCL data which have argued[34,35] that a better way to use information from a continuously monitored SCL is to calculate rates of change in slopes rather than discretely coding SCL events, which discards observations, especially when there is a low sampling rate, as in the present case. In our pilot that validated our SCL facets, the second derivative metric was three times more effective at distinguishing challenge and threat levels to a lab stressor than a frequently-used skin conductance response event coding algorithm (see the SI).

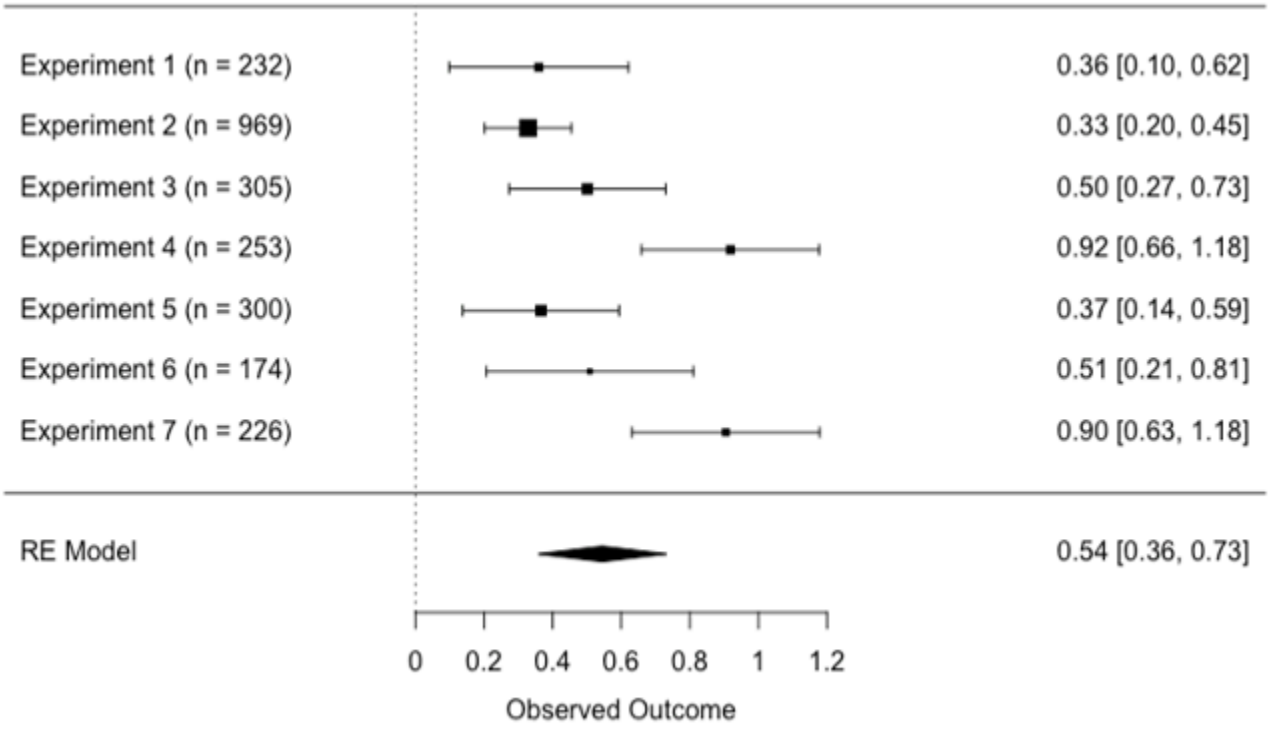



**Figure 4**

**Forest plot for the internal meta-analysis on the key effect of control (i.e., proactive policing) vs. transparency statement condition on civilians' ratings of threat of enforcement for Experiments 1-7.** All analyses come from frequentist statistical methods. *N*'s reflect sample size for key transparent vs. ambiguous condition comparison, rather than the total *N* for each study. Experiment 1: Control *N* = 130, Transparency *N* = 102; Experiment 2: Control *N* = 484, Transparency *N* = 485, Experiment 3: Control *N* = 162, Transparency *N* = 143; Experiment 4: Control *N* = 127, Transparency *N* = 126; Experiment 5: Control *N* = 150, Transparency *N* = 150; Experiment 6: Control *N* = 88, Transparency *N* = 86; Experiment 7: Control *N* = 113, Transparency *N* = 113. Experiments 4-7 only include ambiguous control vs. transparency statement conditions (i.e., Experiment 4 does not include ambiguous transparency condition; Experiment 5 does not include ambiguous transparency condition; Experiment 6 does not include grocery worker conditions; Experiment 7 does not include park ranger conditions). For Experiment 2, we included the focal three-item threat measure that was consistent across all studies, but results are comparable when using the full expanded 10-item measure (see SI). We computed Cohen's *d* and the variance of *d* and then conducted a meta-analysis with the R package *metafor* using a random-effects approach[25]. RE = random effect. Squares show effect size estimates (Cohen's *d*'s). The size of each square gives a representation of each study's sample size. Error bars show 95% confidence intervals (CIs). The diamond represents the point estimate and 95% CI averaged across studies.

# Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

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